Executive Summary

This project investigates the factors influencing student productivity using a dataset of 20,000 student records capturing academic habits, lifestyle factors and digital behaviours. Our initial objective was to model the effect of digital habits, such as phone usage, social media, and gaming, on academic productivity. However, initial Exploratory Data Analysis revealed unusual characteristics including uniformly distributed predictors and a perfect R^2 of 1 in the full Ordinary Least Squares (OLS) model (M_1), suggesting that the dataset may be synthetically generated or structurally engineered. Thus this motivated a shift in focus from purely explaining productivity score to examining the underlying generative process.

A baseline OLS model was first estimated, which produced large t-statistics (more than 10000) and near-zero standard errors, indicating a deterministic relationship between explanatory variables and dependent variables. To obtain a more realistic specification, the significance of a dummy variable *gender* was tested, but was inconclusive. Variables likely to introduce endogeneity such as *final_grade* and *assignments_completed* were removed based on theoretical considerations. Further refinement excluded variables with potential simultaneity concerns, reaching a more realistic model (M_2) at $R^2 = 0.8003$

The project proceeded to examine model assumptions through residual analysis, normality checks and multicollinearity analysis (through variance inflation factors). These found no strong evidence of violations against the linear regression assumptions. However, patterns such as the bounded variable ranges reinforced concerns that the dataset did not reflect real-world stochastic variation.

To complement the OLS analysis, LASSO was employed as a regularization technique to perform automated variable selection together with 10-fold cross validation to the full set of predictors. This identified 6 key predictors: study_hours_per_day, sleep_hours, phone_usage_hours, stress_level, attendance_percentage, and focus_score, the same statistically significant predictors from M_1. The remaining variables were shrunk to zero, corroborating the results from the OLS.

Overall, the findings suggest that the data is likely synthetically generated. While this limits the real-world generalizability of our findings, the internal relationships within the data remain analytically valid and interpretable. Our findings suggest that student productivity is primarily driven by consistent study time, adequate sleep, controlled phone usage, and effective stress management.

Background

With digital devices by our side constantly, we know for a fact that they hinder our productivity be it studying or working. The question we're more interested in is by how much? Regardless, we'll start our analyses from first principles using a dataset containing 20,000 student records, where each observation represents an individual student and their associated academic, behavioral, and lifestyle attributes. The dataset is structured in tabular format, with each column representing a measurable factor potentially influencing the final grades of students.

Our analysis is broken down into four stages. First, we conduct EDA to understand the data structure and surface any anomalies. Second, we estimate a baseline OLS model and iteratively refine it based on causal reasoning. Third, we validate regression assumptions through residual, normality, and multicollinearity diagnostics. Finally, we apply LASSO regularization to confirm variable selection and assess predictive performance. By the end of this project/analysis, we will be able to derive practical insights and provide evidence-based recommendations for improving student productivity.

Exploratory Data Analysis (EDA)

Our dataset contains student records with detailed information about academic habits, lifestyle, digital distraction, and productivity levels. Each row represents one student, and each column represents a specific feature influencing productivity.

First, we load our dataset into RStudio to conduct a preliminary analysis of the data that we'll be working with.

```
'data.frame':   20000 obs. of  18 variables:
 $ student_id           : int  1 2 3 4 5 6 7 8 9 10 ...
 $ age                  : int  23 20 29 27 24 29 21 23 26 19 ...
 $ gender               : Factor w/ 3 levels "Female","Male",..: 1 2 1 1 2 3 1 1 2 1 ...
 $ study_hours_per_day  : num  4.35 6.14 4.98 3.19 7.67 7.18 9.06 6.37 4.19 7.28 ...
 $ sleep_hours          : num  3.63 6.58 3.26 4.58 6.21 3.52 6.36 4.86 4.87 9.56 ...
 $ phone_usage_hours    : num  3.38 5.48 4.83 10.06 3.02 ...
 $ social_media_hours   : num  2.73 1.51 3.63 3.95 1.59 3.74 5.99 1.37 2.87 0.81 ...
 $ youtube_hours        : num  1.83 3.13 0.18 5.75 5.46 1.42 2.2 4.36 0.1 1.35 ...
 $ gaming_hours         : num  5.26 1.73 4.71 2.52 5.65 0.16 4.44 5.13 3.38 2.55 ...
 $ breaks_per_day       : int  6 13 1 9 8 10 14 2 13 7 ...
 $ coffee_intake_mg     : int  347 403 419 178 436 392 87 152 460 416 ...
 $ exercise_minutes     : int  111 28 102 28 105 12 28 103 42 107 ...
 $ assignments_completed: int  2 10 8 18 7 3 15 17 11 6 ...
 $ attendance_percentage: num  57.2 91.3 63.1 40.5 45.5 ...
 $ stress_level         : int  10 10 2 6 6 10 8 6 3 10 ...
 $ focus_score          : int  57 49 38 50 41 70 35 59 39 73 ...
 $ final_grade          : num  81.9 60.9 86.2 71.8 90.1 ...
 $ productivity_score   : num  33.8 49 36.6 19.9 52.9 ...
```

Figure 1: Dataframe of dataset

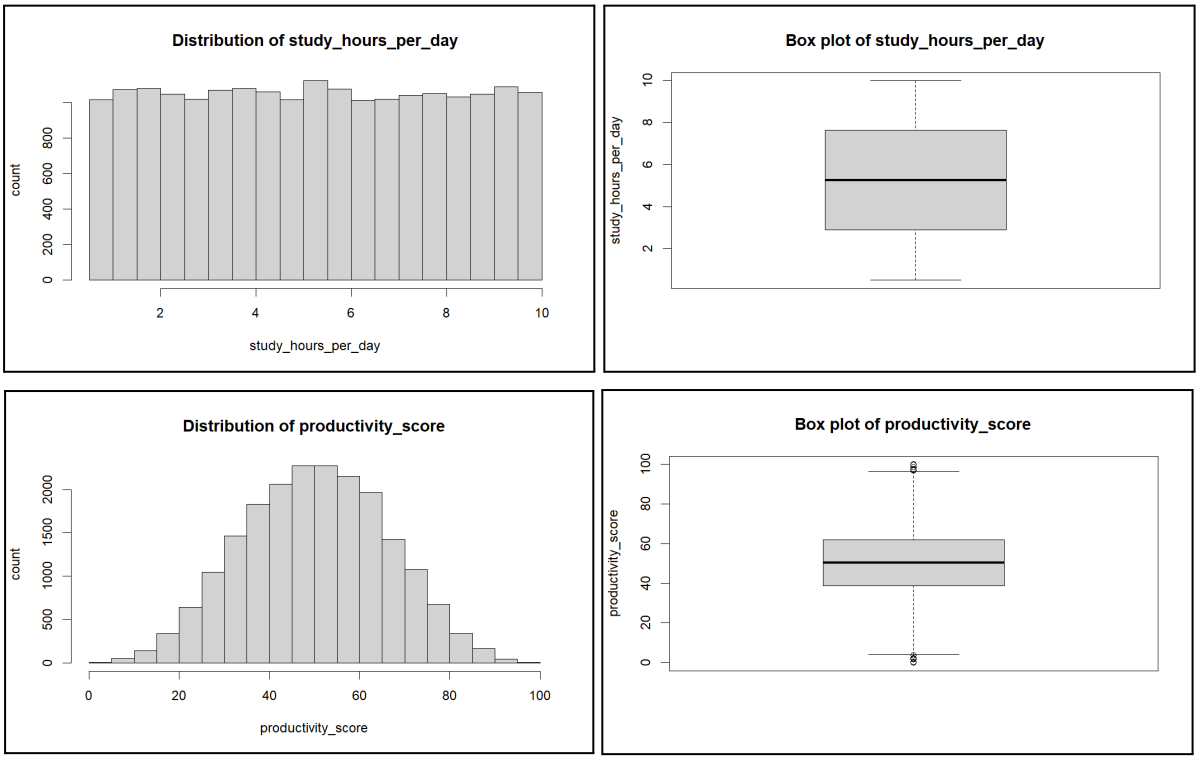
Using the *str()* function, we can see that the dataset is composed of 20,000 rows and 18 columns, along with the data types of the variables. This provides us with an understanding that we'll primarily be working with numerical data, which means that we are likely to build a linear regression model to explain the variance in the students' productivity levels.

Next, we confirm that our dataset has no missing values with *colSums(is.na())* and use *summary()* to obtain descriptive statistics of our variables as shown in Figure 2; looking out for impossible values such as negative number of hours, and checking for skewness to perform data transformations if required.

```
      age            gender      study_hours_per_day  sleep_hours      phone_usage_hours social_media_hours
 Min.   :17.00   Female:9640   Min.   : 0.500      Min.   : 3.000   Min.   : 0.500    Min.   :0.000
 1st Qu.:20.00   Male  :9582   1st Qu.: 2.900      1st Qu.: 4.770   1st Qu.: 3.380    1st Qu.:2.000
 Median :23.00   Other : 778   Median : 5.250      Median : 6.510   Median : 6.240    Median :4.010
 Mean   :23.01                 Mean   : 5.255      Mean   : 6.518   Mean   : 6.250    Mean   :4.004
 3rd Qu.:26.00                 3rd Qu.: 7.640      3rd Qu.: 8.310   3rd Qu.: 9.102    3rd Qu.:5.970
 Max.   :29.00                 Max.   :10.000      Max.   :10.000   Max.   :12.000    Max.   :8.000
 youtube_hours    gaming_hours    breaks_per_day   coffee_intake_mg exercise_minutes assignments_completed
 Min.   :0.000   Min.   :0.000   Min.   : 1.000   Min.   :  0.0    Min.   :  0.00   Min.   : 0.000
 1st Qu.:1.488   1st Qu.:1.490   1st Qu.: 4.000   1st Qu.:125.0    1st Qu.: 30.00   1st Qu.: 4.000
 Median :2.980   Median :2.970   Median : 8.000   Median :249.0    Median : 60.00   Median : 9.000
 Mean   :2.990   Mean   :2.988   Mean   : 7.542   Mean   :249.7    Mean   : 59.65   Mean   : 9.494
 3rd Qu.:4.480   3rd Qu.:4.490   3rd Qu.:11.000   3rd Qu.:373.0    3rd Qu.: 90.00   3rd Qu.:15.000
 Max.   :6.000   Max.   :6.000   Max.   :14.000   Max.   :499.0    Max.   :119.00   Max.   :19.000
 attendance_percentage  stress_level     focus_score     final_grade    productivity_score
 Min.   : 40.00        Min.   : 1.000   Min.   :30.00   Min.   :40.00   Min.   :  0.00
 1st Qu.: 54.81        1st Qu.: 3.000   1st Qu.:47.00   1st Qu.:55.46   1st Qu.: 38.70
 Median : 69.98        Median : 5.000   Median :65.00   Median :70.31   Median : 50.23
 Mean   : 69.95        Mean   : 5.479   Mean   :64.44   Mean   :70.27   Mean   : 50.18
 3rd Qu.: 85.05        3rd Qu.: 8.000   3rd Qu.:82.00   3rd Qu.:85.34   3rd Qu.: 61.78
 Max.   :100.00        Max.   :10.000   Max.   :99.00   Max.   :99.99   Max.   :100.00
```

Figure 2: Summary of dataset

From the results, the mean and median appear to be close for most of our variables, indicating a symmetric distribution. To better assess their spread and identify potential outliers, we used a histogram and boxplot visualization for every numerical variable in the dataset. From the plots, we can see that all the variables are uniformly distributed with no extreme outliers, except for *productivity_score* being normally distributed.



Figures 3-6: Distribution of study_hours_per_day and productivity_score

This unusual finding raised some concerns surrounding our dataset since it did not make sense for many of the variables to have a uniform distribution, especially *sleep_hours*, *final_grade* and *attendance_percentage*. Instead, a dataset dominated by uniform distributions with one normal distribution is suggestive of a potentially simulated, engineered, or highly controlled dataset, rather than a natural phenomenon. Keeping this suspicion at the back of our minds, we check for pairwise correlations between all our numerical variables using a correlation matrix lastly. The purpose is to detect potential multicollinearity in our features and identify strong linear relationships between our features and the target variable.

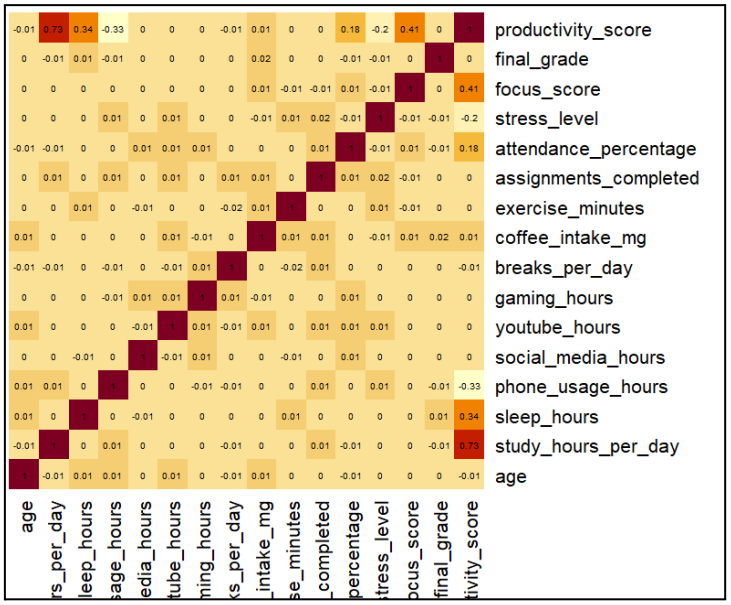


Figure 7: Correlation Matrix of dataset

The heatmap revealed that *study_hours_per_day* had the highest positive correlation to *productivity_score*, followed by *focus_score*, *sleep_hours*, then *attendance_percentage*. Conversely, *phone_usage_hours* and *stress_level* are inversely correlated to *productivity_score*. Apart from these 6 variables however, the rest of the variables had very weak correlation to *productivity_score*.

While these patterns provide initial insights into potential relationships, they are purely correlational and do not account for joint effects or underlying dependencies among variables. As such, they are insufficient for drawing conclusions about the true drivers of productivity. To address this, we proceed to modelling techniques to better understand the key predictors, and assess their relative importance.

Regression Modelling

Baseline Ordinary Least Squares (OLS)

A baseline model using OLS (M_1) approximates the relationship between our explanatory variables and the target variable *productivity_score*, to provide a benchmark for further refining of the model. We obtain the following regression:

```
Coefficients:
                          Estimate Std. Error    t value Pr(>|t|)
(Intercept)           -6.239e+00  2.248e-04 -2.776e+04   <2e-16 ***
age                    2.137e-06  5.405e-06  3.950e-01    0.693
study_hours_per_day    4.316e+00  7.399e-06  5.833e+05   <2e-16 ***
sleep_hours            2.698e+00  9.998e-06  2.698e+05   <2e-16 ***
phone_usage_hours     -1.619e+00  6.126e-06 -2.642e+05   <2e-16 ***
social_media_hours    -2.988e-06  8.803e-06 -3.390e-01    0.734
youtube_hours          1.015e-05  1.173e-05  8.650e-01    0.387
gaming_hours           1.599e-05  1.171e-05  1.365e+00    0.172
breaks_per_day        -2.223e-06  5.054e-06 -4.400e-01    0.660
coffee_intake_mg       2.185e-07  1.412e-07  1.547e+00    0.122
exercise_minutes       7.078e-07  5.864e-07  1.207e+00    0.227
assignments_completed -6.046e-06  3.499e-06 -1.728e+00    0.084 .
attendance_percentage  1.619e-01  1.167e-06  1.387e+05   <2e-16 ***
stress_level          -1.079e+00  7.080e-06 -1.524e+05   <2e-16 ***
focus_score            3.237e-01  1.006e-06  3.219e+05   <2e-16 ***
final_grade           -1.843e-06  1.174e-06 -1.569e+00    0.117
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.002869 on 19984 degrees of freedom
Multiple R-squared:      1,     Adjusted R-squared:      1
F-statistic: 4.191e+10 on 15 and 19984 DF,  p-value: < 2.2e-16
```

Figure 8: Multiple regression with all variables

The estimated coefficients return some unexpected signs. In particular, *gaming_hours*, *youtube_hours* have small positive coefficients. One might expect *gaming_hours* and *youtube_hours* to decrease the *productivity_score* as more entertainment hours tend to leave less time for studying and completing assignments – regarded to be a student's measure of productivity typically. However, the small yet positive coefficients indicate that there is a positive effect on productivity when the number of hours are increased. There is also the possibility that *youtube_hours* could be spent on relevant educational lecture videos that would boost the students' productivity levels, hence the positive sign.

Likewise, one would expect *assignments_completed* and *final_grade* to be positively correlated to *productivity_score*. For variables like *breaks_per_day*, the relationship is ambiguous since it could increase productivity through rest or decrease it with procrastination.

Notably, there are 6 highly significant variables: *study_hours_per_day*, *sleep_hours*, *phone_usage_hours*, *attendance_percentage*, *stress_level* and *final_grade*. *assignments_completed* is also significant at the 10% level. However, the model reports extremely large t-scores (above 10000) due to near 0 standard errors (eg. 0.000007399 for *study_hours_per_day*) alongside a perfect R^2 of 1.

This is highly unusual and combined with the issue of t-scores, It suggests that there are issues such as perfect multicollinearity, functional misspecification or the dependent variable having a deterministic relationship. This prompts further investigation into our model's specification. In particular, some variables may be mechanically related to the dependent variable or may reflect outcomes of productivity rather than its determinants, leading to biased estimates.

To address this, we will re-evaluate the causal relationships between variables and exclude those that are likely to introduce reverse causality or deterministic relationships, with the aim of obtaining a more realistic and interpretable specification for our refined model.

Dummy Variable

Thus far, the dummy variable *gender* has been excluded from the model. The dummy variable *gender* has 3 categories, Female, Male and Other. We now assess its significance to determine whether its inclusion improves the explanatory power of the model and whether it should be kept in the final specification.

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 50.103528   0.163849 305.791   <2e-16 ***
genderMale   0.160763   0.232068   0.693    0.488
genderOther -0.003374   0.599579  -0.006    0.996
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 16.09 on 19997 degrees of freedom
Multiple R-squared:  2.501e-05, Adjusted R-squared:  -7.501e-05
F-statistic:  0.25 on 2 and 19997 DF,  p-value: 0.7788
```

Figure 9: Regressing *productivity_score* against *gender*

The statistical significance of the gender dummy variables was assessed using both individual t-tests and a joint F-test. The estimated coefficients for *genderMale* and *genderOther* are not statistically significant with p values above 0.05. Furthermore, the joint significance F-test fails to reject the null hypothesis that coefficients are equal to zero (p=0.7788). This indicates that gender does not contribute to explaining variation in the dependent variable, consistent with a low R^2 = 0.000025. Therefore, gender is not a significant determinant of productivity_score and can be excluded from the final model specification without the loss of explanatory power.

Refining the Baseline Model

It can be said that *final_grade* is an outcome of productivity such that when productivity increases, the student's final grade is likely to be higher as well. This is defined as reverse causality, where the response variable (*productivity_score*) causes a change in the explanatory variable (*final_grade*). This scenario causes the independent variable to be correlated with the error term in the regression equation, which violates the classical linear regression model assumption that $E(e|X) = 0$ (zero conditional mean). Thus, the variables *final_grade* and *assignments_completed* will be excluded from our model, going by their theoretical relationships.

```
Coefficients:
                       Estimate Std. Error    t value Pr(>|t|)
(Intercept)          -6.239e+00  2.066e-04 -3.020e+04   <2e-16 ***
age                   2.179e-06  5.406e-06  4.030e-01    0.687
study_hours_per_day   4.316e+00  7.399e-06  5.833e+05   <2e-16 ***
sleep_hours           2.698e+00  9.998e-06  2.698e+05   <2e-16 ***
phone_usage_hours    -1.619e+00  6.126e-06 -2.642e+05   <2e-16 ***
social_media_hours   -3.041e-06  8.804e-06 -3.450e-01    0.730
youtube_hours         9.898e-06  1.173e-05  8.440e-01    0.399
gaming_hours          1.594e-05  1.171e-05  1.361e+00    0.174
breaks_per_day       -2.285e-06  5.054e-06 -4.520e-01    0.651
coffee_intake_mg      2.131e-07  1.412e-07  1.509e+00    0.131
exercise_minutes      7.145e-07  5.864e-07  1.218e+00    0.223
attendance_percentage 1.619e-01  1.167e-06  1.387e+05   <2e-16 ***
stress_level         -1.079e+00  7.079e-06 -1.524e+05   <2e-16 ***
focus_score           3.237e-01  1.006e-06  3.218e+05   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.00287 on 19986 degrees of freedom
Multiple R-squared:      1,     Adjusted R-squared:      1
F-statistic: 4.835e+10 on 13 and 19986 DF,  p-value: < 2.2e-16
```

Figure 10: Multiple regression without final_grade and assignments_completed

Dropping the two variables yielded negligible changes, $R^2 = 1$ and t-scores are still inflated. Therefore, the next step would be to remove the variables having a bidirectional relationship with *productivity_score*, namely, *attendance_percentage* and *focus_score*. The relationships of these two variables with productivity are rather ambiguous, whereby attendance may influence productivity while productivity could also be a driving factor for achieving high attendance. Therefore, variables that possess bidirectional relationships with productivity are also excluded due to potential endogeneity concerns.

```
Coefficients:
                     Estimate Std. Error  t value Pr(>|t|)
(Intercept)         26.1513436  0.4438478   58.920   <2e-16 ***
age                 -0.0123435  0.0134443   -0.918    0.359
study_hours_per_day  4.3188654  0.0184031  234.682   <2e-16 ***
sleep_hours          2.6923745  0.0248669  108.271   <2e-16 ***
phone_usage_hours   -1.6213897  0.0152360 -106.418   <2e-16 ***
social_media_hours   0.0079716  0.0218963    0.364    0.716
youtube_hours        0.0240741  0.0291837    0.825    0.409
gaming_hours         0.0126155  0.0291298    0.433    0.665
breaks_per_day       0.0026905  0.0125701    0.214    0.831
coffee_intake_mg     0.0005202  0.0003513    1.481    0.139
exercise_minutes    -0.0008316  0.0014586   -0.570    0.569
stress_level        -1.1018449  0.0176061  -62.583   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 7.137 on 19988 degrees of freedom
Multiple R-squared:  0.8033,    Adjusted R-squared:  0.8032
F-statistic:  7419 on 11 and 19988 DF,  p-value: < 2.2e-16
```

Figure 11: Refined model without final_grade, assignments_completed, focus_score and attendance_percentage

After removing the variables from our regression, we arrive at a refined OLS model (M_2) that better aligns with our intended causal interpretation model. R^2 has decreased to 0.8003, the t-values, albeit high, are now more realistic, and the standard errors for the coefficients seem more reasonable. However, this is still very high for data in behavioural sciences. According to Jacob Cohen $R^2 \geq 0.26$, is a substantial impact within the behavioral sciences.[1] This is

[1] Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Hillsdale, NJ: Lawrence Erlbaum Associates, Publishers.

due to the inherently noisy and heterogenous nature of human behaviour. As such, the high explanatory power observed here may partly reflect the structured or simulated nature of the dataset.

Following this, we will examine whether the remaining explanatory variables satisfy the classical linear regression assumptions. In particular, we need to check for potential nonlinearity, heteroskedasticity, and multicollinearity, given that these assumption violations affect the reliability of the model's coefficient estimates and our statistical inferences as a result.

Model Diagnostics

Non-linearity and Multicollinearity Analysis

Scatter plots were generated to visually examine the relationship between *productivity_score* and four predictors; *study_hours_per_day*, *sleep_hours*, *phone_usage_hours*, and *stress_level*.

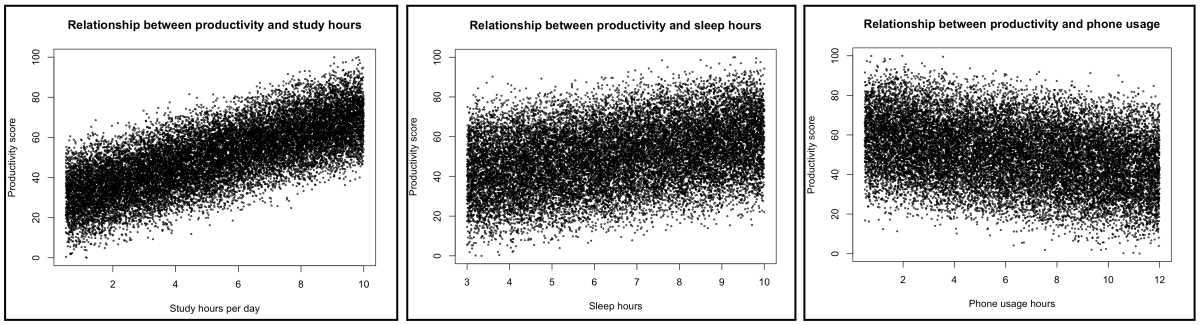
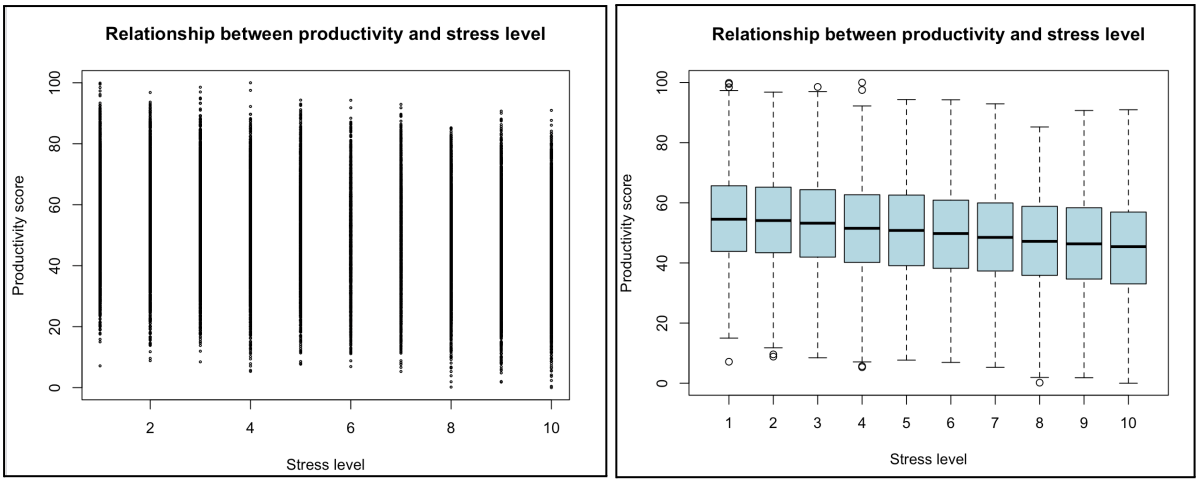


Figure 12-14: Scatter plots of study_hours, sleep_hours and phone_usage_hours

Among the four significant variables, *study_hours* showed the most obvious trend, with a clear upward pattern and a relatively tight cluster of points as seen in *Figure 12*. The other two continuous variables, *sleep_hours* and *phone_usage*, showed weaker trends with much more spread in the data. The number of hours slept has a positive trend, while phone usage has the inverse effect, but in both cases the data points are widely scattered, causing the trends to be less visually convincing on their own.

For stress level, the scatter plot was not very informative since *stress_level* is denoted as an integer on a scale of 1 to 10. By representing the data on a boxplot instead, it enables us to see a pattern almost instantly – the median *productivity_score* noticeably decreases as *stress_level* increases.



Figures 15-16: Scatter plot and Boxplot for stress_level

One thing we noticed across the plots was that the values appear to be cut off at certain boundaries rather than trailing off naturally. For example, the minimum and maximum values of *sleep_hours* seem to be exactly 3 and 10, similar to the other two explanatory variables that seem to stay within very clean ranges. This looks unusual compared to what we would expect from real-world survey or tracking data, where values tend to be less evenly distributed. We will have to continue our analysis before drawing any firm conclusions, but together with the initial R^2 of 1 that we found in M_1, this has added to our concern that the dataset may have been artificially generated. From the scatter plots, none of the relationships show any nonlinear patterns. The trends all appear to be linear without any signs of non-linear patterns, supporting the use of a linear regression model for this data.

Residual Analysis

To further validate the assumptions of the linear regression model, residual plots were generated against the fitted values and each of the four key predictors.

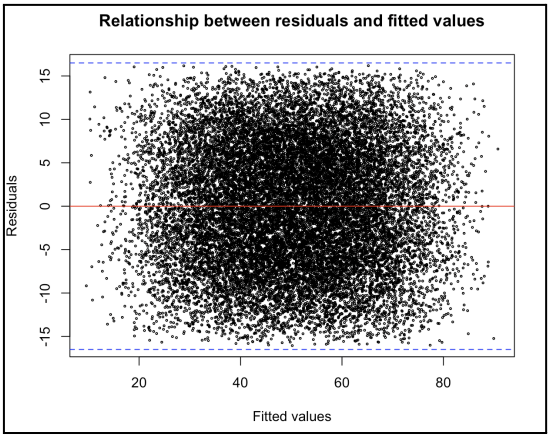
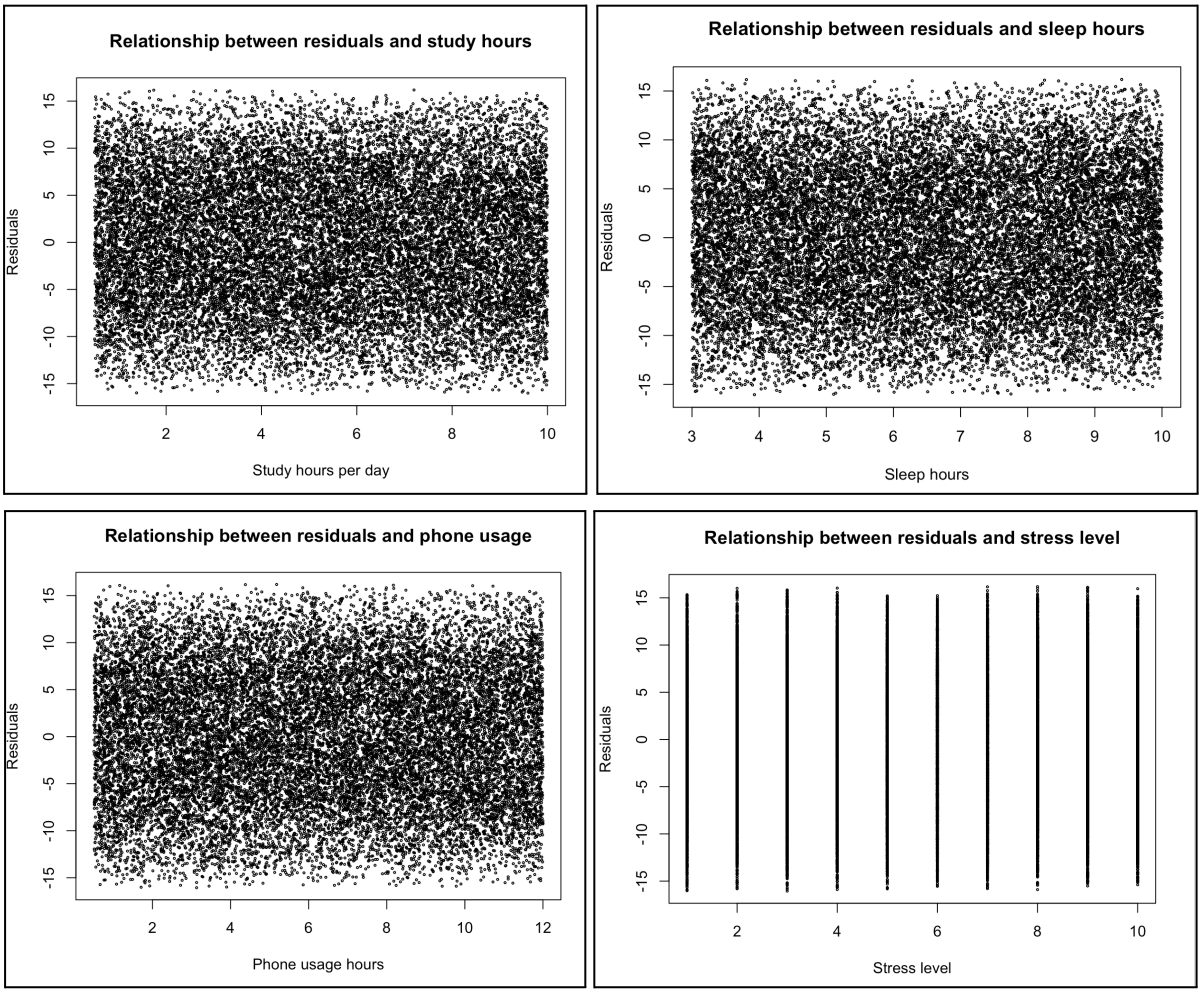


Figure 17: Plot of Residuals against Fitted Values

From the plot above, the residuals appear to be randomly and evenly scattered around zero across the full range of fitted values, with no obvious curved or systematic pattern. This suggests that the linearity assumption holds. The spread of the residuals also stays fairly consistent from left to right, without any clear funneling or widening. This indicates homoscedasticity, meaning the variance of the errors is roughly constant across all levels of the predicted values. Overall, the generated plots support the validity of the linear regression model, as both the linearity and constant variance assumptions are not violated.



Figures 18-22: Residuals vs. Individual Predictors

Once again, the residuals appear to be cut off at certain boundaries rather than trailing off naturally. This is consistent with the clean ranges that we identified earlier in the scatter plots, reinforcing our belief that the dataset may have been artificially generated rather than sampled from actual students.

Normality Analysis on Residuals

Continuing on our model assumptions validation, we ran four diagnostic plots – a histogram, Q-Q plot, CDF plot, and P-P plot – to check if the residuals are normally distributed.

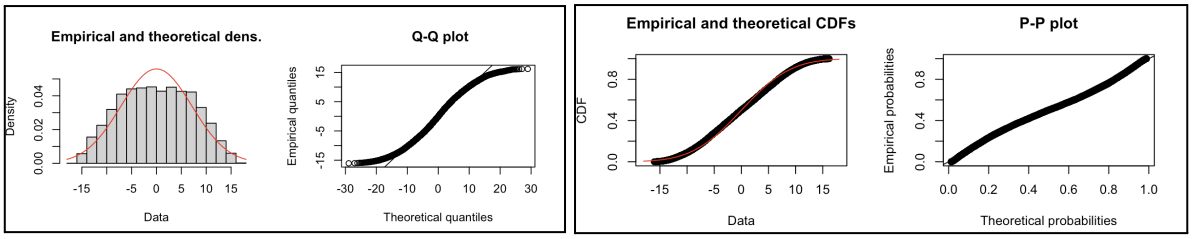


Figure 23: Normality Diagnostic Plots for Residuals

The histogram follows a bell-shaped curve, representing a standard normal distribution while the Q-Q and P-P plots both follow the diagonal line fairly closely. The one thing that stood out was that the centre of the histogram doesn't quite reach the height of the normal curve, so the distribution is a little flatter than what we would expect from a true normal. Given that the diagnostic plots all appear satisfactory regardless, we conclude that the residuals are sufficiently normal for the regression results to remain reliable.

Multicollinearity

```
> vif(M2)
                 age study_hours_per_day          sleep_hours   phone_usage_hours  social_media_hours
            1.000524            1.000346            1.000230            1.000381            1.000224
       youtube_hours       gaming_hours    breaks_per_day    coffee_intake_mg    exercise_minutes
            1.000549            1.000299            1.000626            1.000453            1.000572
        stress_level
            1.000283
```

Figure 24: Variance Inflation Factor (VIF) Output

We used the vif() function from the car package in R to check for multicollinearity among the predictors. All VIF values are very close to 1, which is well below the commonly used thresholds of 5 and 10, indicating that there is virtually no multicollinearity present in the model. Each predictor is providing a unique, independent contribution to explaining the *productivity_score*, resulting in lower standard errors, reliable coefficient estimates, and easier interpretation of individual variable effects.

Regularization

Variable Selection with LASSO

To reverse-engineer the dataset's generative process, LASSO regression is applied to the full set of explanatory variables (excluding only *student_id*, *gender*, *final_grade*, and *assignments_completed*) without the causal pre-filtering applied in M_2. This allows LASSO to determine purely from the data which variables are genuinely relevant, rather than letting our theoretical assumptions potentially bias the model.

The dataset was split into a training set of 16,000 observations and a test set of the remaining 4,000 observations. An OLS baseline model was first fitted on the training set and evaluated on the test set, yielding a test MSE of 8.33×10^{-6} which is an extraordinarily small

value that reflects the near-deterministic nature of the dataset identified earlier. The full model ran on the complete dataset confirms $R^2 = 1$, with highly inflated t-scores for the significant variables (*study_hours_per_day*, *sleep_hours*, *phone_usage_hours*, *attendance_percentage*, *stress_level*, *focus_score*), while the remaining 7 predictors are statistically insignificant at the 5% level.

```
Call:
lm(formula = productivity_score ~ . - student_id - gender - assignments_completed -
    final_grade, data = students)

Residuals:
       Min         1Q     Median         3Q        Max
-0.0051481 -0.0024833 -0.0000161  0.0024697  0.0051084

Coefficients:
                        Estimate Std. Error    t value Pr(>|t|)
(Intercept)           -6.239e+00  2.066e-04 -3.020e+04   <2e-16 ***
age                    2.179e-06  5.406e-06  4.030e-01    0.687
study_hours_per_day    4.316e+00  7.399e-06  5.833e+05   <2e-16 ***
sleep_hours            2.698e+00  9.998e-06  2.698e+05   <2e-16 ***
phone_usage_hours     -1.619e+00  6.126e-06 -2.642e+05   <2e-16 ***
social_media_hours    -3.041e-06  8.804e-06 -3.450e-01    0.730
youtube_hours          9.898e-06  1.173e-05  8.440e-01    0.399
gaming_hours           1.594e-05  1.171e-05  1.361e+00    0.174
breaks_per_day        -2.285e-06  5.054e-06 -4.520e-01    0.651
coffee_intake_mg       2.131e-07  1.412e-07  1.509e+00    0.131
exercise_minutes       7.145e-07  5.864e-07  1.218e+00    0.223
attendance_percentage  1.619e-01  1.167e-06  1.387e+05   <2e-16 ***
stress_level          -1.079e+00  7.079e-06 -1.524e+05   <2e-16 ***
focus_score            3.237e-01  1.006e-06  3.218e+05   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.00287 on 19986 degrees of freedom
Multiple R-squared:      1,     Adjusted R-squared:      1
F-statistic: 4.835e+10 on 13 and 19986 DF,  p-value: < 2.2e-16
```

Figure 25: Multiple regression model with all the predictors

```
> coef(lm.m1)
         (Intercept)                   age   study_hours_per_day           sleep_hours     phone_usage_hours
       -6.238945e+00          2.179225e-06          4.316387e+00          2.697748e+00       -1.618637e+00
  social_media_hours         youtube_hours         gaming_hours        breaks_per_day       coffee_intake_mg
       -3.041224e-06          9.897701e-06          1.593868e-05         -2.284853e-06          2.131022e-07
   exercise_minutes attendance_percentage          stress_level          focus_score
        7.144588e-07          1.618635e-01         -1.079097e+00          3.237288e-01
```

Figure 26: Coefficient estimates of the final model M1

LASSO was then fitted using *cv.glmnet()* with 10-fold cross-validation on the training data, with the optimal regularization parameter selected as the lambda that minimises cross-validation MSE. The optimal lambda was found to be 0.1974, yielding a test MSE of 0.2304. Despite this being larger than the OLS test MSE, it does not come as a surprise — LASSO regularization deliberately introduces bias by penalising coefficients, trading a small amount of predictive accuracy for a sparser, more interpretable model.

After fitting a LASSO regression on the full dataset using the optimal lambda deduced, the following non-zero coefficients were retained:

```
lasso.m2[lasso.m2!=0]
       (Intercept)   study_hours_per_day           sleep_hours     phone_usage_hours attendance_percentage
         -4.5634225             4.2433188             2.6006676            -1.5588965             0.1506632
      stress_level           focus_score
         -1.0114440             0.3141195
```

Figure 27: Coefficient estimates of the final model M_2

7 predictors (*age*, *social_media_hours*, *youtube_hours*, *gaming_hours*, *breaks_per_day*, *coffee_intake_mg*, *exercise_minutes*) had their coefficients shrunk to zero by LASSO, suggesting that they carry no genuine relevance in explaining the response variable (*productivity_score*). Crucially, these are precisely the same variables found to be statistically insignificant in the full OLS model, providing an independent cross-validation of the OLS regression's tests of significance through an entirely different mechanism.

Altogether, LASSO regularization has provided strong evidence of the generative structure of the dataset, identifying the 6 key predictors that collectively account for the near-deterministic relationship observed in M_1.

Predictors	**Effect**	**Interpretation**
study_hours_per_day	+4.243 per hour	Strongest positive driver; each study hour adds about 4.2 points
sleep_hours	+2.601 per hour	Adequate sleep strongly supports productivity
phone_usage_hours	-1.559 per hour	Phone use reduces productivity; each hour costs about 1.6 points
attendance_percentage	+0.151 per unit	Higher attendance is associated with higher productivity
stress_level	-1.011 per unit	Higher stress consistently lowers productivity
focus_score	+0.314 per unit	Higher focus is associated with higher productivity

Table 1: Coefficients and effects of the predictors on productivity score

These findings we have are intuitive and actionable. Interventions targeting study habits, sleep quality, phone use reduction, and stress management are most likely to improve student productivity scores.

Conclusion

This project set out to investigate how digital habits and lifestyle factors influence student productivity, using a dataset of 20,000 student records. Through a structured analytical pipeline spanning EDA, OLS regression, assumption diagnostics, and LASSO regularization, we identified the key drivers of productivity scores and derived actionable recommendations.

Our EDA revealed that the dataset is likely synthetically generated, evidenced by uniformly distributed predictors, clean cut-offs in variable ranges, and an initial R^2 of 1 in the full OLS model. While this limits the real-world generalizability of our findings, the internal relationships within the data remain analytically valid and interpretable.

After refining our OLS model by removing variables exhibiting reverse causality (*final_grade*, *assignments_completed*) and endogeneity (*attendance_percentage*, *focus_score*), we arrived at a more plausible specification with $R^2 = 0.80$. While it remains high by behavioral science standards, it is a significant improvement in realism over the initial deterministic fit. Residual diagnostics confirmed that the assumptions of linearity and homoscedasticity were satisfied, and VIF values indicated no multicollinearity present in the model. Residuals also appear to be normally distributed, completing the set of Gauss–Markov conditions and confirming that our OLS estimators are BLUE under the classical linear regression model assumptions.

To generalize our model, rather than applying LASSO to the causally refined variable set (M_2), we deliberately applied it to the fuller variable set — retaining attendance_percentage and focus_score — to reverse-engineer the dataset's generative process without theoretical pre-filtering. LASSO was fitted with an optimal lambda of 0.1977 via 10-fold cross-validation, yielding a test MSE of 0.2233. Six predictors were retained: *study_hours_per_day* (+4.24 per hour), *sleep_hours* (+2.60 per hour), *phone_usage_hours* (−1.56 per hour), *stress_level* (−1.01 per unit), *attendance_percentage* (+0.151 per unit), and *focus_score* (+0.314 per unit). The seven remaining predictors were shrunk to zero, independently corroborating the OLS findings of insignificance through an entirely different mechanism. The retention of *attendance_percentage* and *focus_score* by LASSO, despite their exclusion from the refined OLS on causal grounds, strongly suggests these variables were embedded in the synthetic generative formula for *productivity_score* rather than being spurious correlates.

Taken together, the findings point to several evidence-based recommendations for improving student productivity: prioritizing consistent study time, ensuring adequate sleep, managing phone use and stress levels, maintaining attendance, and sustaining focus during study sessions. Notably, variables commonly assumed to be distracting such as social media, YouTube, and gaming hours, were found to be statistically insignificant across all models, suggesting their effect on productivity may be overstated or too diffuse to isolate cleanly, even in a controlled synthetic dataset. Future research and tests using real-world survey data would be valuable to validate these relationships outside a synthetic environment.

References

1. Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Hillsdale, NJ: Lawrence Erlbaum Associates, Publishers.
2. *VIF: Variance inflation factors*. RDocumentation. (n.d.). https://www.rdocumentation.org/packages/car/versions/3.1-5/topics/vif